AI HR ATTRITION ANALYTICS

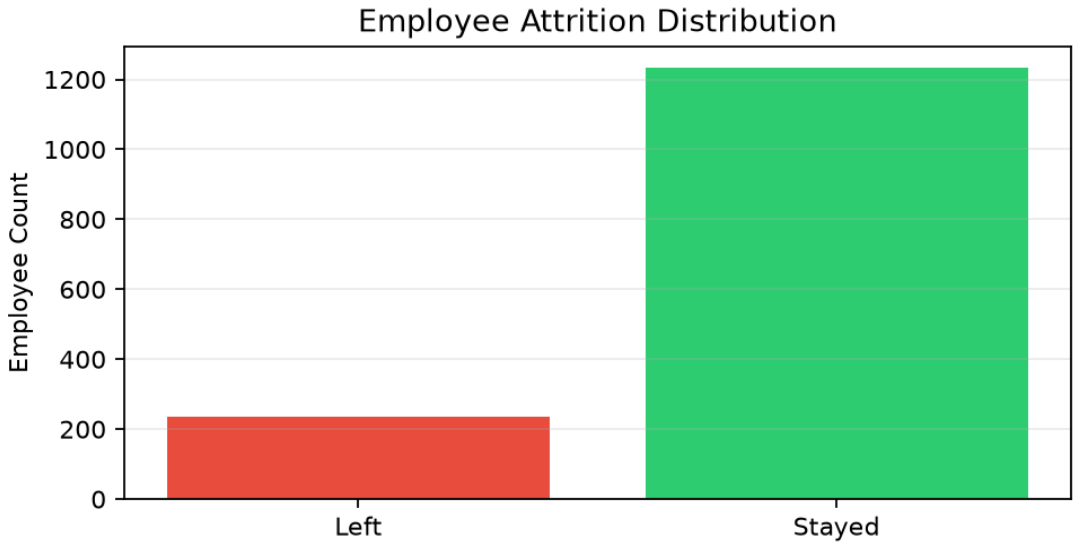
Executive Summary & HR Decision-Support Report

Report Detail	Value
Dataset rows	1,470
Dataset columns	35
Report generated	09 October 2026, 12:45

Purpose: consolidate workforce attrition patterns, root-cause candidates, supporting evidence, model-based scenario results, and proposed HR actions into one report.

1. Executive Summary

Metric	Result
Valid employee records	1,470
Employees who left	237
Employees who stayed	1,233
Observed attrition rate	16.1%



2. Dataset Profile

The selected dataset contains 1,470 rows and 35 columns.

3. Root Cause Candidates & Evidence

Driver	Strength	Difference
Monthly Income	Strong	2045.65
Job Role	Strong	37.26
Age	Strong	26.58
Years at Company	Strong	23.16
Overtime	Strong	20.09
Job Satisfaction	Moderate	11.51
Department	Low	6.79

Driver	Evidence	What the data shows
Monthly Income	Strong	Employees who left had an average monthly income of 4,787.1, compared with 6,832.7 for employees who stayed. The observed difference is 29.9%.

Driver	Evidence	What the data shows
Job Role	Strong	Sales Representative shows the highest observed attrition rate at 39.8%, compared with Research Director at 2.5%. The observed difference is approximately 37.3 percentage points.
Age	Strong	18-25 shows the highest observed attrition rate at 35.8%, compared with 36-45 at 9.2%. The observed difference is approximately 26.6 percentage points.
Years at Company	Strong	0-2 Years shows the highest observed attrition rate at 29.8%, compared with 11-20 Years at 6.7%. The observed difference is approximately 23.2 percentage points.
Overtime	Strong	Yes shows the highest observed attrition rate at 30.5%, compared with No at 10.4%. The observed difference is approximately 20.1 percentage points.
Job Satisfaction	Moderate	1.0 shows the highest observed attrition rate at 22.8%, compared with 4.0 at 11.3%. The observed difference is approximately 11.5 percentage points.
Department	Mixed	Sales shows the highest observed attrition rate at 20.6%, compared with Research & Development at 13.8%. The observed difference is approximately 6.8 percentage points.

Interpretation: observed associations are useful for investigation, but they do not establish that a factor causes employee attrition.

4. What-If Analysis

Selected driver: Overtime

Scenario Metric	Value
Current Probabilities	[88.89809924 19.60776792 83.32291319 ... 64.20059117 11.07222529 42.30378311]
Scenario Probabilities	[69.33441992 19.60776792 58.51895222 ... 33.61513277 11.07222529 42.30378311]
Current Categories	['High' 'Low' 'High' ... 'Medium' 'Low' 'Medium']
Scenario Categories	['Medium' 'Low' 'Medium' ... 'Low' 'Low' 'Medium']
Current High	206
Scenario High	102
Current Average	38.383423896499195
Scenario Average	32.07671261439581
Risk Change	6.306711282103386
High Risk Reduction	104
Auc	0.8061848565767938

Scenario results are model-based estimates, not proof of causal impact or guaranteed future outcomes.

5. Recommendations & Action Plan

Driver	Priority	Focus Group	Focus Employees	Focus Attrition Rate	Recommended Action	Suggested Owner	Suggested Timeframe	Success Metric	Proposed Target Rate
Age	High	(17.999, 30.0]	386	25.9	Review career development, onboarding, flexibility, and employee feedback across age groups.	HR	30–90 days	Retention and development participation by group	23.3
Job Role	High	Sales Representative	83	39.8	Review role-specific workload, career progression, training, and compensation.	HR and Functional Managers	30–90 days	Role-level attrition and internal mobility	35.8
Monthly Income	High	(1008.999, 2911.0]	369	29.3	Review compensation fairness by role, experience, responsibility, and relevant market benchmarks.	HR and Compensation Team	30–90 days	Pay-equity indicators and subsequent attrition	26.4
Overtime	High	Yes	416	30.5	Review workload, overtime frequency, staffing, and employee feedback.	HR and Department Managers	30–60 days	Overtime frequency and subsequent attrition rate	27.4
Years at Company	High	(-0.001, 3.0]	470	26.0	Review onboarding, recognition, career progression, and support across tenure groups.	HR and People Managers	30–90 days	Retention by tenure group and internal promotions	23.4
Job Satisfaction	Medium	1 — Lowest satisfaction	289	22.8	Investigate satisfaction concerns involving management, workload, recognition, and career development.	HR and People Managers	30–60 days	Satisfaction score and subsequent attrition rate	20.5
Department	Monitor	Sales	446	20.6	Compare staffing, workload, management practices, and development opportunities across departments.	HR Business Partner	30–60 days	Department-level attrition and employee feedback	18.5

Proposed targets are planning goals and should be reviewed against actual outcomes. They are not guaranteed results.

6. Limitations & Responsible Use

• The report reflects the selected dataset and the analysis available at generation time.

• Missing data and small group sizes can affect the stability of findings.

• Associations do not establish causation.

• Model predictions can be uncertain and should be validated.

• HR decisions should consider business context, privacy, fairness, and employee feedback.

End of Report